

Differential Case

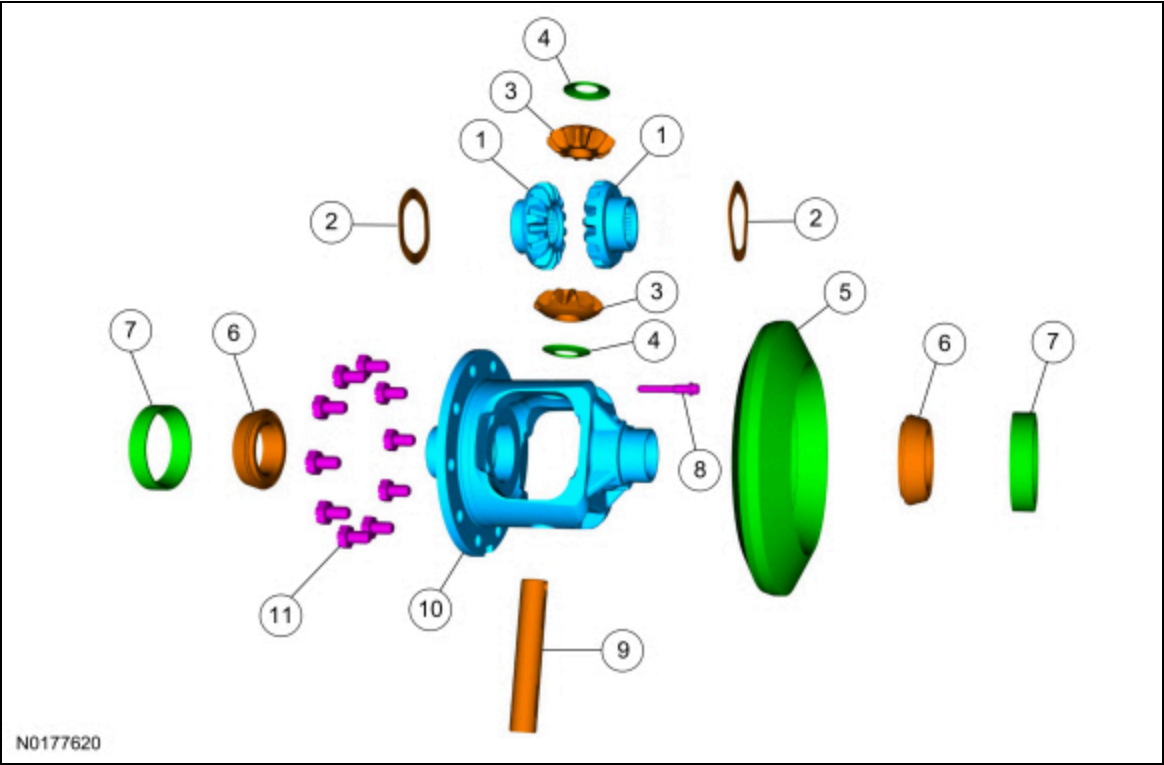
 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST1749-A	Gauge, Differential (Traction-Lock) 205-386 (T97T-4205-D)
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Material

Item	Specification
Motorcraft® SAE 80W-90 Premium Rear Axle Lubricant (US); Motorcraft® SAE 80W-90 Premium Axle Lubricant (Canada) XY-80W90-QL (US); CXY-80W90-1L (Canada)	WSP-M2C197-A
Motorcraft® Threadlock and Sealer TA-25-B	WSK-M2G351-A5



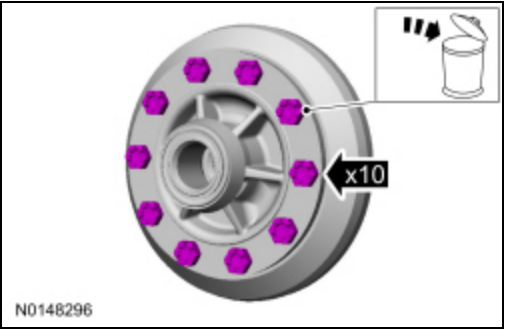
Item	Part Number	Description
1	4236 4236	Differential side gears
2	—	Differential side gear wave washers
3	4215 4215	Differential pinion gears

4	4230 4230	Differential pinion thrust washers
5	4209 4209	Differential ring gear
6	4221 4221	Differential bearings (2 required)
7	4222 4222	Differential bearing cups (2 required)
8	4241 4241	Differential pinion shaft lock bolt
9	4211 4211	Differential pinion shaft
10	4204 4204	Differential case
11	4216 4216	Differential ring gear bolt (10 required)

Disassembly

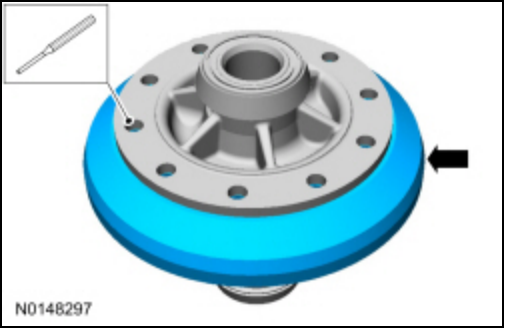
NOTE: *The Torsen® differential should be replaced as a unit only. Do not attempt to rebuild.*

1. Remove the differential carrier. Refer to Differential Carrier.
2. Remove and discard the differential ring gear bolts.

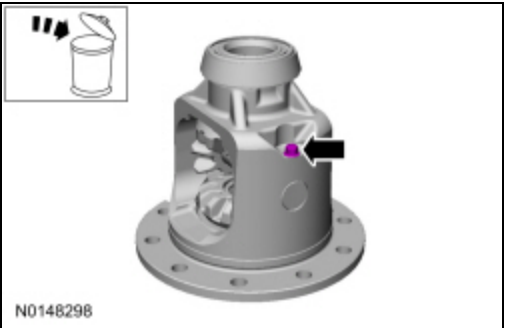


3. **NOTICE:** Care should be taken not to damage the differential ring gear bolt hole threads.

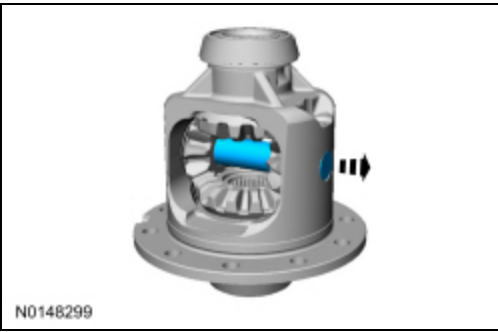
Using a drift punch separate the ring gear from the differential case.
 General Equipment: Drift Punch



4. Remove and discard the differential pinion shaft bolt.

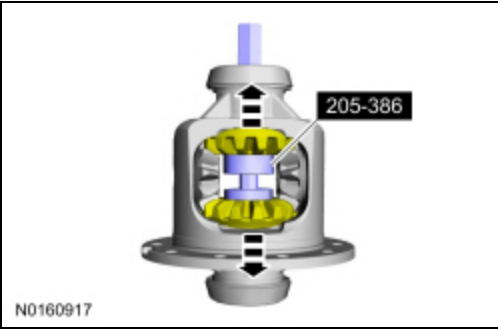


5. Remove the differential pinion shaft.



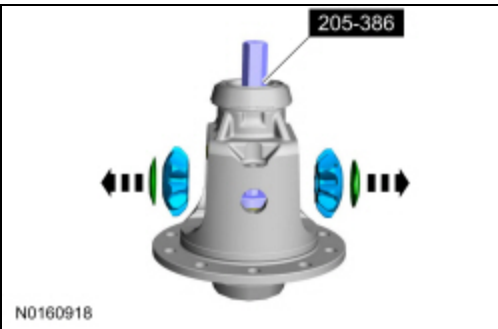
6. **NOTE:** *Do not over tighten the special tool.*

Using the special tool, compress the differential side gear wave washers.



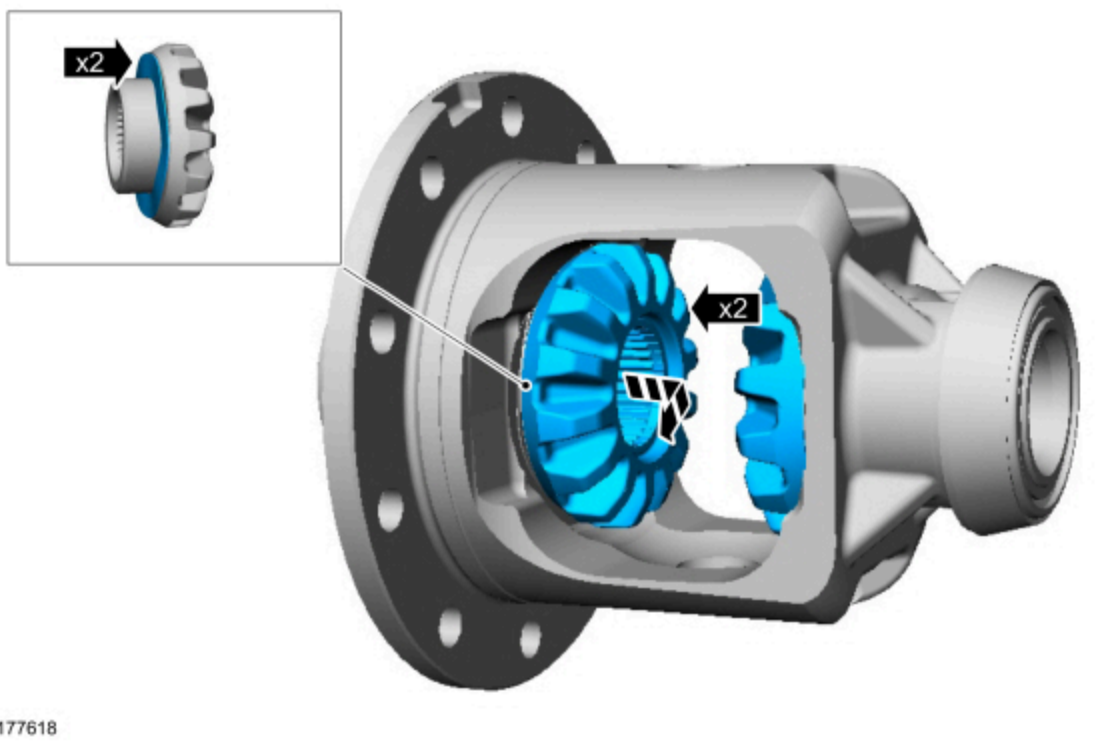
7. Remove the differential pinion thrust washers and the differential pinion gears.

1. Rotate the differential pinion gears to the differential case window, and remove them.
2. Separate the thrust washers from the gears.



8. **NOTE:** *Note the position of the differential side gear wave washer for reassembly.*

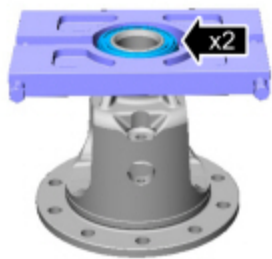
Remove the differential side gear assemblies.



9. Using the general equipment, remove and discard the differential side bearings.

General Equipment: Hydraulic Press

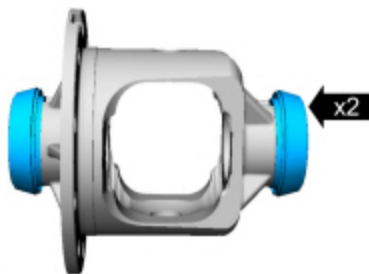
General Equipment: Bearing Puller



Assembly

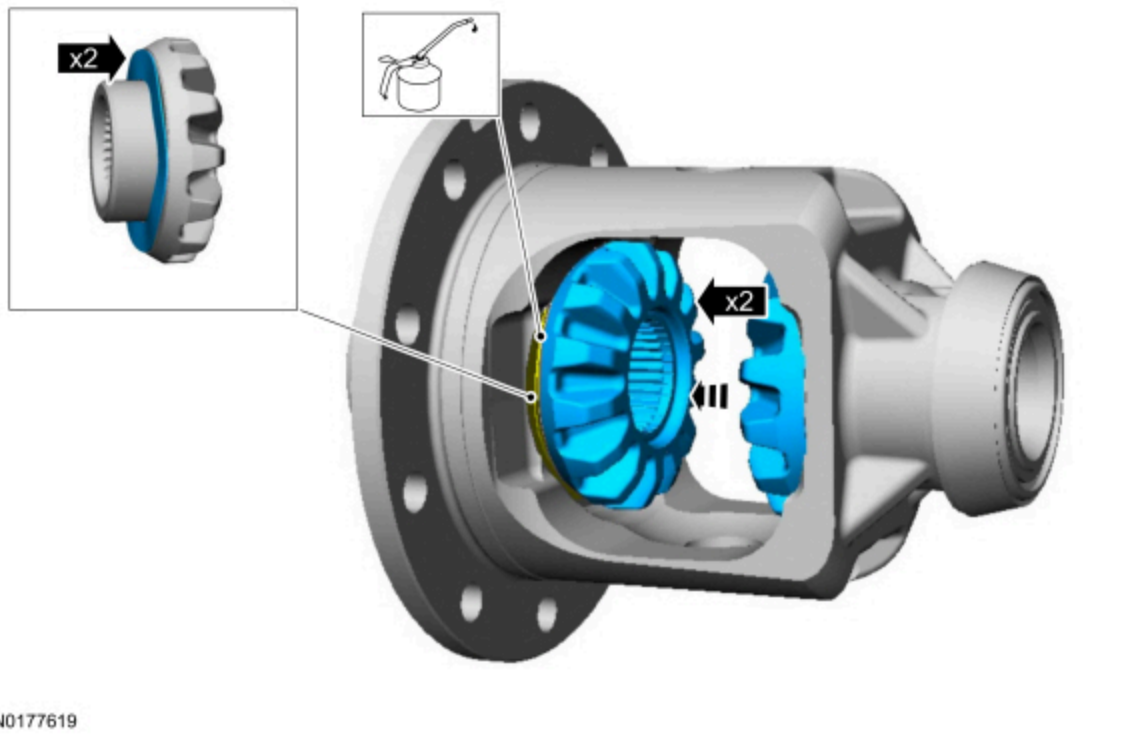
1. Using general equipment install the new differential side bearings.

General Equipment: Hydraulic Press

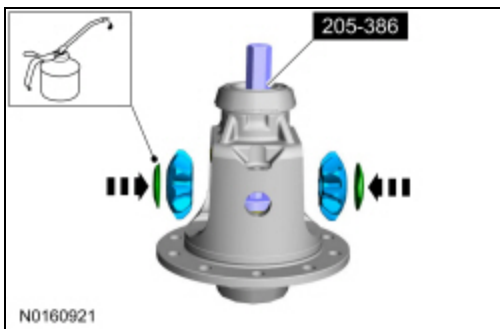


2. Install the differential side gears in the differential carrier.

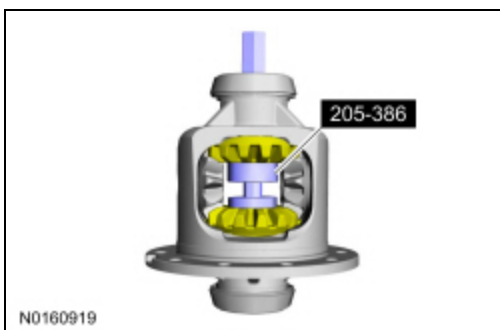
- Lubricate the differential wave washer with axle lubricant.
- Install the differential wave washer on the differential side gears.
- Install the differential side gear assemblies into the differential carrier.



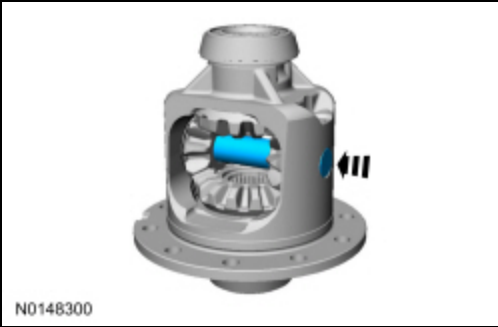
3. With the special tool installed, lubricate the differential pinion thrust washers and the differential pinion gears with axle lubricant and assemble them.



4. Using the special tool, rotate the differential pinion gears to align with the differential pinion shaft bore and remove the special tool.



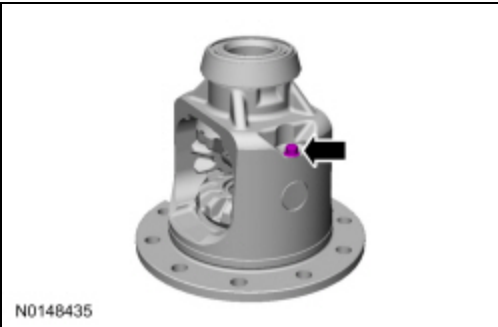
5. Install the differential pinion shaft.



6. **NOTE:** *If a new pinion shaft bolt is unavailable, coat the threads with sealer.*

Install the new differential pinion shaft bolt.

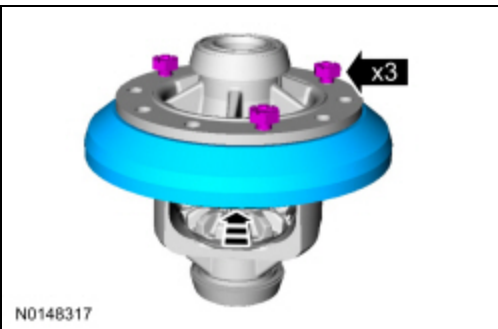
- Tighten to 30 Nm (22 lb-ft).



7. Install the differential ring gear.

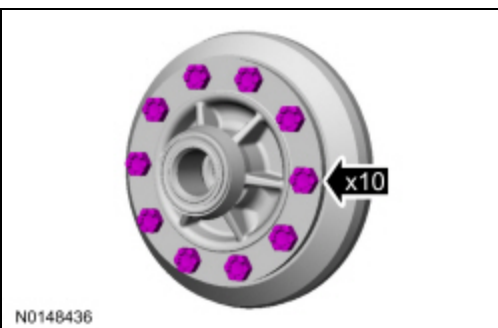
General Equipment: Hydraulic press

1. Place the differential ring gear on the differential assembly.
2. Hand-start 3 ring gear bolts to align the holes in the differential ring gear and the differential assembly.
3. Place the differential carrier and differential ring gear onto the press bed blocks with the differential ring gear teeth facing downward.
4. Press the differential ring gear into place.



8. Install the 7 remaining new bolts.

- Tighten bolts to 128 Nm (95 lb-ft).



9. Install the differential carrier. Refer to Differential Carrier.